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Beyond genes: EpiSwitch[®] and Orion platform-powered 3D genome architecture biomarkers reveal shared biology across ME/CFS, long COVID, PTSD, rheumatoid arthritis, and multiple sclerosis

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Abstract

Background Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS), Long COVID (LC19), post-traumatic stress disorder (PTSD), rheumatoid arthritis (RA), and multiple sclerosis (MS) are clinically distinct disorders that share substantial symptom overlap, including persistent fatigue, cognitive impairment, autonomic dysfunction, and immune dysregulation. Although these conditions differ in diagnosis and clinical presentation, their underlying biological mechanisms remain poorly understood and may involve convergent regulatory pathways.

Methods The EpiSwitch[®] 3D genomics platform and Orion knowledgebase were used to integrate chromosome conformation signatures with genome-wide association study (GWAS)-derived datasets across ME/CFS, LC19, PTSD, RA, and MS. Three-dimensional genomic anchors were mapped to coding genes and analysed using STRING protein–protein interaction networks and Cytoscape-based systems biology approaches. Disease-specific anchor datasets were generated and compared at both gene and network levels to identify shared biological processes and regulatory mechanisms.

Results Analysis of the ME/CFS dataset identified 552 unique 3D genomic anchors mapped to 567 genes, with analogous disease-specific anchor sets generated for LC19, PTSD, RA, and MS. Direct overlap between disease-associated genes was limited; however, higher-order network analyses revealed substantial interconnectivity and convergence across conditions. Shared biological pathways included immune and cytokine signalling, interferon responses, mitochondrial function, metabolic regulation, and neuroendocrine processes. Highly connected hub genes included immune regulatory nodes such as *LAG3* and components of the mTOR signalling pathway,

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implicating T-cell exhaustion, chronic immune activation, and immunometabolic dysregulation as common mechanisms underlying these disorders.

Conclusions These findings support a systems-level model in which clinically overlapping fatigue-associated syndromes arise from perturbations of interconnected regulatory networks rather than discrete disease-specific pathways. Despite limited genetic overlap, substantial convergence at the network level suggests shared biological architecture across ME/CFS, LC19, PTSD, RA, and MS. The identification of common regulatory pathways provides a mechanistic framework for the development of cross-disease diagnostic and therapeutic strategies. By capturing dynamic regulatory states, 3D genomic biomarkers offer significant potential for objective blood-based diagnostics, patient stratification, and the identification of shared therapeutic targets across complex chronic disorders. These findings support the application of precision medicine approaches and may accelerate the development of novel interventions for fatigue-associated multisystem diseases.

Keywords Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS), Long COVID, Post-traumatic stress disorder, Multiple sclerosis, Rheumatoid arthritis, 3D genomics, Chromosome conformation capture, Chromatin architecture, Epigenetic biomarkers, Systems biology, Network biology, Protein–protein interaction networks, Biomarker discovery, Precision medicine

Introduction

Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) is a complex, debilitating, and heterogeneous disorder characterised by profound fatigue, post-exertional malaise, cognitive dysfunction, sleep disturbance, and a range of autonomic and immune abnormalities (reviewed in [1]). Despite decades of research, the pathophysiology of ME/CFS remains incompletely understood, and no universally accepted diagnostic biomarker has been established [2]. Diagnosis continues to rely largely on symptom-based clinical criteria, resulting in delays, misclassification, and significant unmet clinical need [3]. This lack of objective diagnostics has contributed to historical under-recognition of the disease and has impeded the development of effective therapeutics [4].

Recent advances in molecular biology, particularly in genomics and epigenetics, have begun to reshape understanding of ME/CFS as a biologically grounded, multi-system disorder. Increasingly, evidence supports the view that ME/CFS arises from dysregulation across interconnected biological systems, including the immune, metabolic, neuroendocrine, and neurological domains [1]. However, translating these insights into clinically actionable tools has proven challenging, in part due to the heterogeneity of findings across studies and the limitations of traditional molecular approaches [5].

Current understanding of ME/CFS pathophysiology

A central challenge in ME/CFS research is its multifactorial and heterogeneous nature [6]. Multiple, non-mutually exclusive mechanisms have been proposed, including immune dysregulation, mitochondrial dysfunction, autonomic imbalance, and neuroinflammation [7–10]. These mechanisms likely interact in a dynamic and patient-specific manner.

Immune dysfunction has emerged as one of the most consistently reported features of ME/CFS [7, 11]. Numerous studies have identified abnormalities in cytokine signalling, T-cell activation, and innate immune responses [12–14]. In particular, dysregulation of interleukin pathways, including IL-2, IL-6, and IL-10, has been implicated in disease pathogenesis [15, 16]. These alterations may reflect chronic immune activation or impaired immune resolution following an initial trigger, such as viral infection [17].

The role of post-infectious mechanisms is particularly relevant, as many patients report disease onset following infections such as Epstein–Barr virus or, more recently, SARS-CoV-2 [17, 18]. These observations have led to the hypothesis that ME/CFS may represent a maladaptive response to infection, involving persistent immune activation and altered host–pathogen interactions [1]. Epigenetic changes induced by infection may contribute to this process by reprogramming gene expression patterns in immune cells [19, 20].

Mitochondrial dysfunction and altered energy metabolism represent another key area of investigation (reviewed in [21]). Patients with ME/CFS frequently exhibit impaired oxidative phosphorylation, altered glycolysis, and reduced capacity for energy production [22]. These abnormalities are thought to underlie the hallmark symptom of post-exertional malaise, in which even minor physical or cognitive exertion leads to prolonged symptom exacerbation [23]. Importantly, metabolic changes are closely linked to immune function, suggesting a coordinated dysregulation of immunometabolic pathways [7].

Neuroendocrine and autonomic dysfunction also play important roles. Alterations in hypothalamic–pituitary–adrenal (HPA) axis function, including reduced cortisol responsiveness, have been reported [24]. Epigenetic changes in genes such as NR3C1, which encodes the

glucocorticoid receptor, provide a potential mechanistic link between stress response pathways and disease manifestations [25]. These findings are complemented by evidence of autonomic nervous system imbalance, including orthostatic intolerance and heart rate variability abnormalities [26].

Collectively, these observations support a model in which ME/CFS arises from dysregulation across multiple interacting biological systems, rather than a single causative pathway [1, 6–10]. However, identifying consistent biomarkers within this complex landscape has remained a major challenge [27].

Proposed pathophysiological mechanisms in long COVID and PTSD

In parallel with advances in ME/CFS research, increasing attention has focused on the biological mechanisms underlying Long COVID (LC19) and post-traumatic stress disorder (PTSD), both of which share overlapping clinical features and may involve convergent molecular pathways. Although these conditions differ in their primary triggers (viral infection in Long COVID and psychological trauma in PTSD), emerging evidence suggests convergence on common dysfunction across immune, metabolic, and neuroendocrine systems [28, 29].

Long COVID is characterised by persistent symptoms following SARS-CoV-2 infection, including fatigue, cognitive impairment, and autonomic dysfunction, closely resembling ME/CFS [26]. Proposed mechanisms include persistent immune activation, with elevated cytokines and dysregulated interferon responses, potentially driven by residual viral antigens [28]. Autoimmunity has also been reported, with autoantibodies targeting receptors involved in autonomic and vascular regulation [30]. In addition, endothelial dysfunction and microvascular injury may contribute to impaired tissue oxygenation and chronic symptoms [31]. Metabolic abnormalities are increasingly recognised, including mitochondrial dysfunction and altered energy metabolism, which may underlie fatigue and post-exertional malaise [32]. These changes are closely linked to immune pathways, supporting an integrated immunometabolic model. Neurobiological mechanisms, including neuroinflammation and autonomic nervous system imbalance, further contribute to cognitive and systemic symptoms [33].

PTSD, while traditionally viewed as a psychiatric disorder, is also associated with systemic biological changes [34]. Central features include dysregulation of the hypothalamic–pituitary–adrenal (HPA) axis, leading to altered cortisol responses and impaired stress adaptation. Structural and functional changes in brain regions involved in fear processing, along with altered neurotransmitter signalling, contribute to core symptoms [35]. Importantly, PTSD is also linked to chronic inflammation and

immune activation, with evidence of stress-induced epigenetic modifications affecting immune gene regulation [36]. This highlights a bidirectional interaction between the neuroendocrine and immune systems.

Despite distinct initiating factors, Long COVID and PTSD share key features with ME/CFS, including immune dysregulation, metabolic disturbance, and neuroinflammation. These overlaps suggest that these conditions may represent different manifestations of a shared underlying network dysfunction, supporting the need for integrative approaches, such as 3D genomic analysis, to elucidate common mechanisms.

Limitations of traditional genetic and molecular approaches

Genome-wide association studies (GWAS) have provided important insights into the genetic architecture of many complex diseases. However, in ME/CFS, GWAS findings have been relatively modest, with only a limited number of reproducible loci identified [37, 38]. These loci are typically associated with immune and neurological pathways, but do not fully explain disease heterogeneity or clinical presentation. One limitation of GWAS is its focus on linear DNA sequence variation, which captures only a fraction of the functional genome [39]. The majority of disease-associated variants lie within non-coding regions, where their functional impact is mediated through regulatory mechanisms rather than changes in protein sequence. These regulatory effects are highly context-dependent and often involve interactions between distant genomic elements [40]. Furthermore, genetic variation alone does not account for environmental influences, which are particularly relevant in all three conditions. Factors such as infection, stress, and environmental exposures can profoundly influence disease onset and progression, yet their effects are not directly captured by DNA sequence analysis.

Other molecular approaches, including transcriptomics, proteomics, metabolomics, and microbiome analyses, have identified a range of abnormalities in ME/CFS [4]. For example, proteomic studies have suggested associations with antigen-driven B-cell expansion [41], while metabolomic analyses have identified disruptions in energy metabolism [42]. However, these approaches often yield inconsistent results across cohorts and lack sufficient diagnostic accuracy for clinical use [2].

Indeed, multiple candidate diagnostic modalities have been proposed, including extracellular vesicle profiling, Raman spectroscopy of peripheral blood mononuclear cells, metabolomics, and neuroimaging [2, 43]. While some of these approaches have demonstrated moderate accuracy (e.g., ~70–90%), they are often limited by technical complexity, lack of reproducibility, or challenges in clinical implementation. These limitations highlight

the need for integrative approaches that capture the dynamic, multidimensional nature of disease biology [1].

Epigenetics as a central mechanism in ME/CFS

Epigenetics refers to changes in gene expression that do not involve alterations in the underlying DNA sequence. These changes include DNA methylation, histone modifications, non-coding RNA regulation, and higher-order chromatin organisation [44]. Epigenetic mechanisms provide a critical interface between genetic predisposition and environmental influences, making them particularly relevant for diseases such as ME/CFS, LC19 and PTSD [45].

A growing body of evidence supports the role of epigenetic dysregulation in ME/CFS [9, 46]. Genome-wide methylation studies have identified thousands of differentially methylated positions across genes involved in immune function, metabolism, and neuroendocrine regulation [19]. These changes are not random but are enriched in pathways consistent with known disease mechanisms.

Importantly, epigenetic modifications are dynamic and can change in response to environmental stimuli [47]. For example, infection-induced epigenetic changes may alter the expression of immune-related genes, leading to persistent dysregulation even after the initial trigger has resolved [48]. Similarly, stress-related epigenetic changes may affect neuroendocrine pathways, contributing to symptom persistence [49].

Three-dimensional genome architecture and disease

Recent advances in genomics have revealed that the genome is organised in a highly structured three-dimensional (3D) architecture within the nucleus [50]. This organisation is not random but plays a critical role in regulating gene expression by bringing distal regulatory elements, such as enhancers and promoters, into physical proximity. Chromosome conformation (CC) capture technologies have enabled the study of these spatial interactions, revealing networks of regulatory loops that control gene activity [51].

We have developed a novel epigenetic assay (EpiSwitch® Explorer Assay), a bespoke design based on Agilent SurePrint 1 M array, that allows simultaneous screening for 10⁶ of 3D chromosomal conformations in the circulating blood cells [52]. Using EpiSwitch® technology, we have identified epigenetic signatures correlating to metabolic conditions and neuroinflammatory diseases such as amyotrophic lateral sclerosis (ALS) [53] and inflammatory conditions such as rheumatoid arthritis (RA) [54]. We have further identified chromosome conformations specific to the response to PD-1/PD-L1 immune therapy [55]. Interestingly, although the 3D genomic regulatory architecture encompasses the whole genome, by mapping

the top 3D genomic biomarkers to the closest genetic loci captured by their topological control (within 3Kb), it is possible to broaden the biological insights into genes, pathways and protein networks under the influence of 3D genomic regulation and associated cellular phenotype, contributing to the pathology of a disease and identify potential therapeutic strategies.

In a recent study, we employed the EpiSwitch® Explorer array platform and machine learning algorithms to predict individual responses to COVID-19 infection [56]. We developed a blood-based prognostic test to assess infection severity and identified 3D genomic markers linked to potential treatments in pathways relevant to immune function, including T-cell signalling, macrophage-stimulating protein (MSP)-RON signalling, and calcium signalling. EpiSwitch®-based commercial tests are now available to diagnose prostate cancer with 94% accuracy (PSE test) [57] and response to immune checkpoint inhibitors across 14 cancers with 85% accuracy (CiRT test) [55].

Development of 3D genomic biomarkers for ME/CFS

In our recent study, we used whole-blood samples from patients with severe ME/CFS and healthy controls and a genome-wide 3D genomic screening approach (EpiSwitch® platform) to identify diagnostic biomarkers for ME/CFS [58]. Machine-learning-based feature selection identified a panel of 200 chromosome conformation markers that distinguished ME/CFS patients from controls with 96% accuracy. Importantly, these markers were distributed across the genome, supporting a polygenic and network-based model of disease. Pathway analysis of the associated genes revealed strong enrichment in immune and inflammatory signalling pathways, including interleukin signalling (IL-2), TNF pathways, and JAK/STAT signalling, supporting previous findings [12, 59].

IL-2 is a critical regulator of T-cell function, and its dysregulation may contribute to both immune activation and immune exhaustion [60, 61]. The identification of IL-2-associated markers also enabled the stratification of patients into subgroups, suggesting potential for personalised therapeutic approaches [58]. Furthermore, the overlap between ME/CFS biomarker networks and therapeutic pathways, including those targeted by drugs such as rituximab and glatiramer acetate, highlights the potential for 3D genomic analysis to inform treatment strategies [62, 63].

Conclusion/rationale: The clinical potential of 3D genomics in ME/CFS

The transition from symptom-based diagnosis to objective, blood-based biomarkers represents a paradigm shift for ME/CFS and related conditions. By capturing functional regulatory changes through epigenetic and 3D

genomic signatures, clinicians can achieve greater diagnostic stability and scalability than traditional methods allow.

Beyond diagnosis, integrating these 3D genomic insights with other “omics” layers facilitates:

- **Precision Medicine:** Better patient stratification and targeted immunomodulatory therapies.
- **Mechanistic Clarity:** Identifying shared regulatory networks across ME/CFS, Long COVID, and PTSD.
- **Drug Discovery:** Pinpointing common therapeutic targets within complex molecular pathways.

Ultimately, leveraging advanced platforms such as EpiSwitch® advances the field toward a more integrated, mechanistic understanding of these disorders, paving the way for faster diagnoses and more effective, personalised treatment strategies.

Methods

Study design

The objective of this study was to investigate whether distinct chronic conditions characterised by persistent fatigue exhibit convergent 3D genomic regulatory architecture. Rather than treating these disorders as a single biological entity, we integrated disease-specific EpiSwitch® and GWAS-derived datasets to identify shared regulatory networks and candidate molecular nodes that may provide a framework for understanding biological convergence across fatigue-associated conditions.

The diseases included in this analysis were selected because persistent fatigue is a clinically important feature across these conditions and because they encompass diverse inflammatory, immune-mediated, neurological and stress-related disease contexts. The analysis therefore uses disease-specific molecular datasets as a means of investigating potential convergence across conditions characterised by fatigue, rather than directly testing fatigue as an independent trans-disease phenotype.

The analytical workflow comprised three components: (i) incorporation of previously characterised and independently validated ME/CFS chromosome conformation signatures and diagnostic classifier data from Hunter et al. (2025) [58]; (ii) derivation of 3D genomic anchors from GWAS datasets for ME/CFS, Long COVID (LC19), post-traumatic stress disorder (PTSD), multiple sclerosis (MS) and rheumatoid arthritis (RA) using EpiSwitch® Orion; and (iii) integration of these datasets into network-based analyses using EpiSwitch® Orion analytical tools, STRING and Cytoscape for functional interpretation and identification of candidate hub genes.

Clinical cohorts and sample collection

Clinical samples for the ME/CFS diagnostic biomarker analysis were derived from the previously characterised retrospective case–control study reported in [58]. That study included a discovery cohort of 47 patients with severe ME/CFS and 61 age-matched healthy controls, together with a separate independent validation cohort comprising 24 ME/CFS cases and 45 controls. The 24 ME/CFS and 45 control samples were held out from model development and were used specifically to independently evaluate the previously developed 200-marker EpiSwitch® ME/CFS diagnostic classifier, which achieved 92% sensitivity, 98% specificity and 96% overall accuracy [58]. Inclusion criteria for ME/CFS patients included: diagnosis according to established clinical criteria, age 20–80 years, and severe disease status. Exclusion criteria included chronic comorbidities, malignancy, autoimmune disease, and treatment with DNA-modifying or biological therapies. Control subjects were age-matched individuals without a history of chronic illness or key ME/CFS symptoms. All participants provided written informed consent. The study was approved by the relevant ethics committees and conducted in accordance with the Declaration of Helsinki and Good Clinical Practice guidelines [58].

Importantly, the independent validation cohort was used to assess the diagnostic classifier and was not used to derive or independently validate the network hub genes reported in the present study. The current study uses the previously characterised ME/CFS EpiSwitch® biomarker dataset as an input to a subsequent integrative network analysis with EpiSwitch® Orion-derived GWAS anchor datasets from ME/CFS, Long COVID, PTSD, multiple sclerosis and rheumatoid arthritis. Consequently, the hub genes identified in the present study should be considered network-derived candidate regulatory nodes requiring independent experimental and clinical validation.

Chromosome conformation capture workflow and EpiSwitch® 3D genomic profiling

Chromosome conformation capture was performed using the EpiSwitch® platform, based on a modified 3C methodology. The detailed protocol was described previously [58]. Briefly, fixed chromatin was digested with restriction enzymes, followed by proximity ligation to join spatially adjacent DNA fragments. Crosslinks were reversed, and ligation products were amplified and quantified (Fig. 1). Genome-wide interrogation of chromatin interactions was performed using high-density EpiSwitch® Explorer microarrays (Agilent SurePrint platform), enabling analysis of approximately 1 million potential interaction sites across the genome. Array hybridisation and data acquisition were conducted according to established protocols.

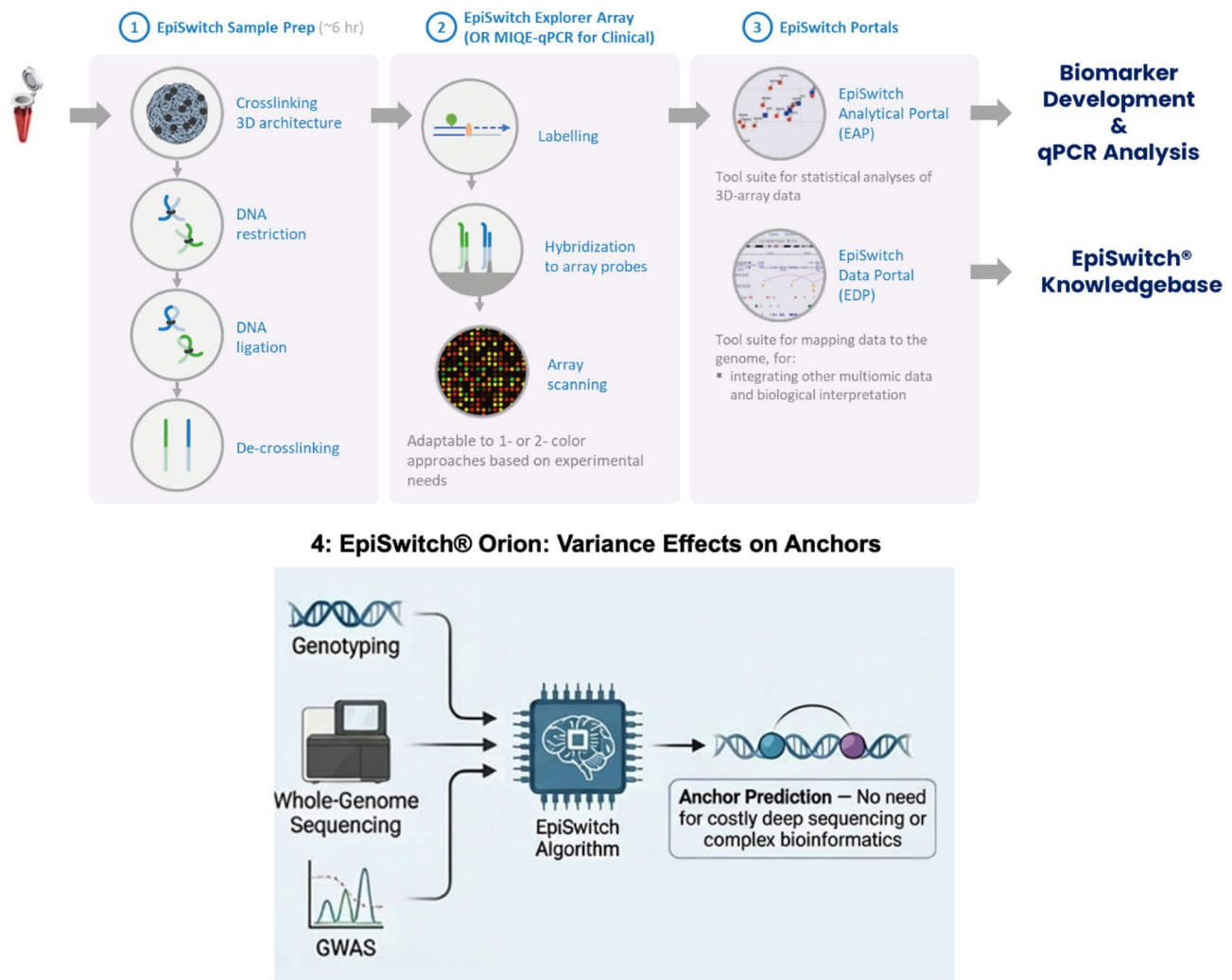


Fig. 1 The EpiSwitch® 3D Genomics Platform. (1) Sample Preparation: Whole blood samples undergo in situ formaldehyde crosslinking to fix 3D chromatin loops, followed by restriction digestion and proximity ligation to create chimeric DNA fragments representing genomic interactions. (2) Digital Detection: After reverse-crosslinking, these fragments are analysed via high-density EpiSwitch® Explorer arrays or qPCR to generate a binary library of chromosome conformation signatures (CCS). (3) Data Analytics: The resulting signatures are processed through the EpiSwitch® Analytical and Data Portals for feature selection and genomic mapping. Validated 3D signatures are integrated into the EpiSwitch® knowledgebase, which combines proprietary 3D data with global multi-omic datasets (GWAS, RNA-seq, proteomics, biological pathways and semantically parsed content from publication, in a large language model used for data mining) (4) EpiSwitch®Orion: In silico determination of the effects of genomic variance on the genomic anchor landscape — the sites where genomic looping can occur — using the EpiSwitch anchor algorithm coupled with variance information from WGS, GWAS, or genotype data

Microarray data processing and feature selection

Raw microarray data were subjected to background correction and quantile normalisation using the EpiSwitch® analytical pipeline (R-based). Batch effects were corrected using ComBat. The dataset was partitioned into training and test subsets prior to analysis to prevent data leakage. Differential chromatin interactions between ME/CFS cases and controls were identified using both parametric (linear modelling via limma) and non-parametric (Rank Product) approaches. Markers were filtered based on statistical significance ($p \leq 0.01$) and abundance score thresholds. The intersection of markers identified by both methods was used for downstream modelling. Recursive

feature elimination was applied to derive a panel of the top 200 CCS markers associated with disease status.

Machine learning and classifier development

Predictive modelling was performed using gradient boosting (XGBoost) implemented in R. Hyperparameters were optimised using grid search, with regularisation and subsampling applied to minimise overfitting. Model interpretability was assessed using SHapley Additive exPlanations (SHAP) values. Model performance was evaluated using sensitivity, specificity, positive predictive value (PPV), negative predictive value (NPV), and overall

accuracy. Independent validation was conducted on a held-out cohort to assess generalisability.

GWAS data acquisition and processing

GWAS summary statistics for ME/CFS, PTSD, and Long COVID were obtained from publicly available datasets and consortium-based studies (Table 1). These datasets included large-scale population-level genetic data, enabling the identification of disease-associated loci.

The following GWAS datasets were used:

LC: <https://www.ebi.ac.uk/gwas/studies/GCST90454540>;

PTSD:<https://pgc.unc.edu/for-researchers/download-results/>;

ME/CFS:<https://www.medrxiv.org/content/https://doi.org/10.1101/2025.08.06.25333109v1>;

RA:<https://www.ebi.ac.uk/gwas/studies/GCST90132223>;

MS: <https://www.ebi.ac.uk/gwas/studies/GCST003566>.

The GWAS SNPs were filtered at a significance threshold of $p \leq 0.01$, and the retained variants were processed through EpiSwitch Orion v1 anchor detection; only anchors exceeding a probability of 0.99 were carried forward for further analysis. Protein–protein interaction

networks for the resulting anchor genes were constructed in STRING using the default medium-confidence score cutoff (0.4), and the networks were imported into Cytoscape (version 3.10.3) for visualisation and topological analysis. Pathway enrichment was assessed with a false discovery rate (FDR) multiple-testing correction.

For each condition, anchors were generated de novo. GWAS summary statistics were filtered as described above, and the resultant SNPs/indels were used as input within the EpiSwitch® Orion workflow. Each effect variant was compared against the reference genome (hg38); where the effect allele differed from the reference allele, the variant replaced the reference allele prior to anchor calling. The EpiSwitch anchor detection algorithm was then run independently across each condition/patient dataset to generate the anchor landscape for that condition. Each disease-specific anchor landscape was compared to the reference genome (hg38) anchor landscape to identify anchors that were novel to disease (i.e., not present in the reference landscape) and anchors present in the reference but absent in disease; Fig. 4A shows anchors present only in disease relative to the reference. The Venn diagrams in Figs. 3A and 4A compare these disease-specific anchors after mapping to their nearest gene

Table 1 Summary of GWAS datasets used in this study

Trait / GWAS	Reference	Study design & ancestry composition	Cases (n)	Controls (n)	Total N	Notes
Rheumatoid arthritis	Ishigaki et al. 2022 Nat Genet 54 [11]:1640–1651 https://doi.org/10.1038/s41588-022-01213-w	Multi-ancestry meta-analysis 37 cohorts across 5 ancestries: EUR (25), EAS (8), AFR (2), SAS (1), ARB (1)	35,871	240,149	276,020	Ancestry breakdown (cases): EUR 22,350; EAS 11,025; AFR 999; SAS 986; ARB 511. Control total as reported in main text (not broken down by ancestry in source table/Fig. 1a).
PTSD	Nievergelt et al. 2024 Nat Genet 56 [5]:792–808 PGC-PTSD Freeze 3	Multi-ancestry meta-analysis European (EA), African/ admixed African (AA), Native American/Latin (LAT)	150,793	1,130,197	1,280,990	Effective N (n_{eff}) by ancestry: EA 641,533; AA 42,804; LAT 6,530 (not summed for combined analysis).
Multiple sclerosis	Andlauer et al. 2016 Sci Adv 2:e1501678	German discovery (DE1 + DE2, pooled) with Sardinian replication External eQTL/mQTL reference panels: MPIP, GTP, GTEx	7,791	13,718	21,509	Discovery (DE1 + DE2): 4,888 cases / 10,395 controls. Sardinian replication: 2,903 cases / 3,323 controls. Figures combine discovery + replication case–control cohorts only.
Long COVID	Lammi et al. 2025 Nat Genet 57 [6]:1402–1417 PMID 40,399,555	24 discovery meta-analyses, 16 countries, 6 genetic ancestries Replication: 8 cohorts + VA Million Veteran Program	15,950	1,892,830	1,908,780	Overall study totals (discovery + replication combined). Primary discovery model (strict case/broad control, 11 studies): 3,018 cases / 994,582 controls; FOXP4 (rs9367106) genome-wide significant, OR 1.63.
ME/CFS	DecodeME study	GWAS-1 (primary), case–control European ancestry cases vs. UK Biobank controls	15,579	259,909	275,488	Cases = 93.1% of QC-passed cohort (16,730). Five additional sensitivity GWAS (female/male/ no-infection/infection-stratified/GWAS-2) and external replication cohorts (R-1, R-2) also reported.

Sample sizes reflect the primary/headline discovery (and, where stated, combined discovery + replication) figures reported in each source publication. Where a study reports multiple case/control definitions or sub-analyses, the headline discovery figures are given in the main columns, and secondary figures are summarised under Notes

(see “Mapping of 3D Anchors to Genes” above), a deliberate choice made to allow biologically meaningful comparison across conditions at the level of affected genes/pathways rather than at the level of raw anchor coordinates, since anchor coordinates are condition-specific and are not expected to overlap precisely across datasets even where the same underlying gene is implicated.

EpiSwitch® Orion platform translates linear GWAS variant data into functional three-dimensional genomic architecture. Orion maps millions of non-coding genetic variants (single nucleotide polymorphisms (SNPs)) to chromatin anchors that regulate the genome’s ability to fold and organize in 3D space. This enables the creation of a comprehensive “atlas” identifying which genetic variations actively influence disease mechanisms through anchor interactions and chromatin looping.

For each condition:

- ME/CFS datasets (including DecodeME) were processed to generate 552 unique 3D anchors mapped to 567 genes.
- PTSD datasets yielded 362 anchors mapped to 324 genes.
- Long COVID datasets were processed using analogous workflows to generate disease-specific anchor sets. 611 anchors mapped to 567 genes.
- MS 730 anchors mapped to 629 genes.
- RA 965 anchors mapped to 885 genes.

These anchors represent genomic regions involved in spatial interactions and are hypothesised to influence gene regulation.

Mapping of 3D anchors to genes

3D genomic anchors were mapped to their nearest coding genes based on genomic proximity and regulatory context. This mapping process incorporated additional data sources, including enhancer–promoter interactions, expression quantitative trait loci (eQTLs), and transcription factor binding sites. The resulting gene sets were used as inputs for downstream network and pathway analyses. Importantly, mapping was not restricted to linear proximity but also considered spatial relationships inferred from chromatin interaction data. Anchor-to-gene assignment itself was performed using bedtools closest, which reports the nearest gene(s) to each anchor by genomic distance. Where two or more genes are equidistant from a given anchor (a tie), bedtools closest reports all tied genes by default rather than arbitrarily selecting one, which produces occasional one-to-many anchor-to-gene relationships; conversely, a single gene can be the nearest match for more than one anchor when that gene spans, or lies near, several anchor loci. Anchor and gene counts are therefore not expected to be equal

(e.g., ME/CFS: 552 anchors → 567 genes; PTSD: 362 anchors → 324 genes), and this reflects the gene-annotation step rather than an error in anchor calling.

Network construction and integration

Gene sets derived from CCS markers and GWAS anchors were analysed using STRING to construct protein–protein interaction (PPI) networks. Networks were visualised and integrated using Cytoscape. Disease-specific networks were merged to generate a composite interaction network. Topological properties, including degree centrality, betweenness centrality, and clustering coefficients, were calculated to identify highly connected hub nodes.

Protein–protein interaction network construction

Gene sets derived from CCS markers and GWAS anchors were analysed using STRING (Search Tool for the Retrieval of Interacting Genes/Proteins) to construct protein–protein interaction (PPI) networks. Default STRING parameters were used, incorporating both experimentally validated and predicted interactions. Networks were generated separately for each condition gene sets. Networks were visualised and further analysed using Cytoscape software. Nodes represent proteins encoded by genes, while edges represent functional associations, including direct physical interactions and indirect functional relationships.

Network integration and topological analysis

Individual disease-specific networks were merged to create a composite network representing all five conditions. Nodes were colour-coded according to disease origin, enabling visualisation of overlap and interconnectivity. Topological properties of the merged network were analysed, including: Degree centrality: Number of connections per node, Betweenness centrality: Measure of node importance in connecting network regions and Clustering coefficient: Degree of local interconnectedness. Highly connected nodes (hub genes) were identified based on degree centrality. Subsets of top nodes (e.g., top 50 or top 200) were extracted for detailed analysis.

Pathway enrichment and functional analysis

Pathway enrichment analysis was performed using gene sets mapped from CCS markers and GWAS anchors. Analyses incorporated multiple databases, including Reactome, KEGG, Gene Ontology, and MSigDB. Significantly enriched pathways were identified using hypergeometric testing with multiple-comparison correction. Functional clustering was applied to group related pathways and identify dominant biological themes.

Gene prioritisation and network propagation, clustering and subgroup analysis

Network-based gene prioritisation was performed using a Random Walk with Restart (RWR) algorithm. Known disease-associated genes were used as seed nodes, and candidate genes (genes under investigation) were ranked based on RWR score (gives higher scores to candidate nodes that are strongly connected to the guide nodes). Hierarchical clustering of CCS profiles was performed to identify potential molecular subgroups within the ME/CFS cohort. Clustering was based on similarity in chromatin interaction patterns, with emphasis on markers associated with key immune pathways.

Statistical analysis

All statistical analyses were performed in R (version 4.2.0). Power calculations indicated that the study was adequately powered to detect significant differences between groups. Statistical significance was defined as $p \leq 0.05$ unless otherwise specified.

Results

Overview of analytical workflow and dataset integration

A detailed STRING network analysis of the top 200 EpiSwitch® ME/CFS biomarkers identified in the whole-genome screening [58] provides insight into disease-specific regulatory architecture. Figure 2A illustrates a densely connected network, with nodes representing genes mapped from chromosome conformation signatures and edges indicating functional associations. The network exhibits clear clustering patterns, with groups of genes forming modules corresponding to biological pathways. These clusters are not randomly distributed but instead reflect coordinated functional processes. Prominent hubs include IL-2, IL-10, TNF α , and TLR-related proteins, reflecting perturbations in cytokine signalling, innate immune sensing, and downstream JAK/STAT and NF- κ B cascades. The density of connections within clusters suggests strong functional coherence, while inter-cluster links indicate crosstalk between biological processes. The overall structure of the network is highly integrated, with few isolated nodes. This reflects the interconnected nature of the biological systems underlying ME/CFS and supports the concept of a multi-system disorder. Importantly, the network includes both genes previously associated with ME/CFS and additional candidate genes identified through EpiSwitch® analysis that, to our knowledge, have not previously been associated with ME/CFS. These findings illustrate how 3D genomic approaches can expand the range of candidate disease-associated genes by identifying regulatory relationships that may not be apparent from conventional linear genomic analyses.

Pathway enrichment analysis of ME/CFS-associated genes reveals significant clustering within key biological pathways. Figure 2 panels B and C present Reactome and NCI & Single Cell Marker pathway clusters derived from the top 100 highly connected nodes within the ME/CFS network. We observe enrichment in several major biological domains within the Reactome pathways: Immune and inflammatory signalling, including cytokine pathways such as IL-1 β and TNF- α , Interferon signalling: indicative of antiviral responses, Mitochondrial function (including mitochondrial biogenesis) and metabolism (specifically glycolysis and fatty acid oxidation): including pathways related to energy production, Cellular stress response: reflecting adaptation to physiological stressors, Signalling to NTRKs, Transcriptional regulation by MECP2 and FOXO, VEGF signalling, and Neurotransmission: including glutamate and GABA signalling. Furthermore, analysis of the NCI & Single Cell Marker pathways revealed significant clustering in PDGFR-beta Signalling, the Glucocorticoid receptor network, CXCR4-mediated signalling, S1P1-S1P3 pathways, Hedgehog signalling, ER-alpha signalling, P38 MAPK signalling, TRK receptor signalling, SMAD2/3 signalling, and the ATF-2 transcription network. The clustering of these pathways suggests that ME/CFS involves coordinated dysregulation across multiple biological systems. Notably, many of these pathways are also implicated in Long COVID and PTSD [28, 30, 34, 35], providing a mechanistic basis for the network interconnectivity observed in earlier figures. The prominence of immune and metabolic pathways aligns with clinical observations of immune activation and energy dysfunction in ME/CFS patients. Similarly, the involvement of neurotransmission pathways supports the neurological symptoms associated with the condition.

To explore specific mechanistic insights, network-based disease prediction analysis was performed. Figure 2D highlights the integration of the immune checkpoint gene LAG3 into the ME/CFS network. In this figure, known ME/CFS-associated genes (guide genes) are represented alongside candidate genes identified through network association. LAG3 appears as a strongly connected node within this network, linked to the ME/CFS prior-knowledge space, despite not being part of the original guide gene set. The positioning of LAG3 within the network indicates close functional relationships with established ME/CFS genes. Its connectivity suggests a role in the underlying disease biology, potentially through modulation of immune responses, particularly in T-cell exhaustion following chronic immune activation.

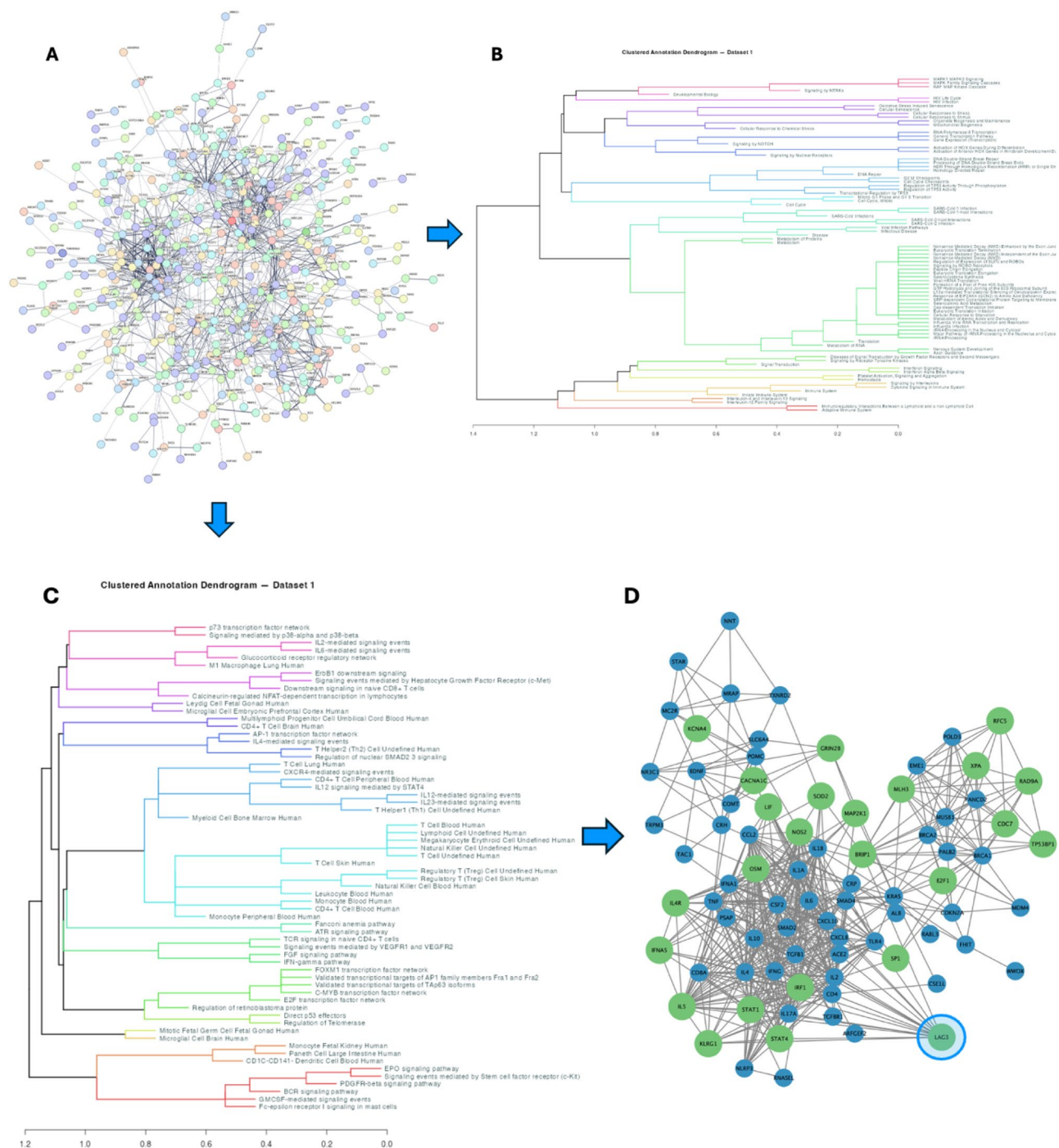


Fig. 2 STRING Network of ME/CFS Biomarker Genes Reveals Functional Clustering. **(A)** STRING network of the top 200 EpiSwitch ME/CFS model markers mapped to the closest coding region. **(B)** Reactome pathways clustered on the basis of the gene membership from the top 100 nodes (based on the degree of connection in the network). Most notable pathways include: Signalling to NTRKs, Transcriptional regulation by MECP2 and FOXO, VEGF signalling, Mitochondrial biogenesis, Cytokine signalling (IL-1beta, TNF-alpha, etc.), Cellular response to stress, Interferon signalling, Metabolism (glycolysis and fatty acid oxidation), and Altered neurotransmission (Glutamate, GABA). **(C)** NCI & Single Cell Marker pathways clustered on the basis of the gene membership from the top 100 nodes (based on the degree of connection in the network). Most notable pathways include: PDGFR-beta Signalling, Glucocorticoid receptor network, CXCR4-mediated signalling, S1P1-S1P3 pathways, Hedgehog signalling, ER-alpha signalling, P38 MAPK signalling, TRK receptor signalling, SMAD2/3 signalling, and ATF-2 transcription network. **(D)** Gene Disease Prediction via Network Associations. This predicts candidate genes for complex diseases by utilising a curated functional gene network and the Random Walk with Restart (RWR) algorithm. It shows the marker-associated gene LAG3 linked to the ME/CFS prior-knowledge space. The positioning of LAG3 indicates a functional relationship with established ME/CFS genes. Given that LAG3 is a recognised immune checkpoint molecule associated with T-cell exhaustion, its network position provides a potential link between ME/CFS-associated regulatory networks and altered T-cell responses during chronic immune activation. Representative network visualisations; complete gene and pathway datasets are provided in Supplementary Tables 1–3

3D Genomic analysis reveals high network interconnectivity despite low gene-level overlap in ME/CFS, long COVID, and PTSD

Application of the EpiSwitch® Orion platform enabled the transformation of linear genomic variation into three-dimensional (3D) regulatory insight across ME/CFS, Long COVID (LC19), and PTSD datasets. Disease-specific chromosome conformation signatures (CCS, loops) derived from the clinical cohorts and EpiSwitch® anchors derived from GWAS (Table 1) summary statistics were mapped to coding regions, producing gene sets for each condition. These gene sets were subsequently analysed through network biology approaches, including STRING protein–protein interaction modelling and Cytoscape-based integration.

An Initial comparison of gene sets derived from GWAS-based 3D anchors demonstrated minimal overlap between ME/CFS, Long COVID, and PTSD. The Venn diagram (Fig. 3A) illustrates that the majority of genes identified in each condition are unique, with only a small subset shared across two or more diseases. Specifically, each disease exhibited a substantial number of exclusive genes following anchor-to-gene mapping. ME/CFS gene sets, derived from 552 3D genomic anchors mapping to 567 genes, showed limited direct intersection with PTSD (324 genes) and Long COVID (567 genes) gene sets. The overlapping regions in the Venn diagram are visually small relative to the total area of each circle, emphasising the low degree of shared gene identity. This finding is consistent with prior GWAS studies of complex diseases, in which heterogeneity and polygenic architecture lead to diffuse and condition-specific gene associations [64].

To investigate functional relationships between disease-associated genes, STRING protein–protein interaction networks were constructed for each condition and subsequently merged using Cytoscape. Figure 3B presents the resulting integrated network, with nodes spatially separated by disease origin and colour-coded accordingly. In contrast to the sparse overlap observed in Fig. 3A, the merged network (Fig. 3B) reveals a densely interconnected structure. Nodes representing genes from ME/CFS, Long COVID, and PTSD are not isolated within discrete clusters but instead exhibit extensive cross-linking through shared interaction partners. The resulting network demonstrates multiple bridging connections between disease-specific clusters, forming a continuous interaction landscape. These interconnections suggest that although different genes may be implicated in each condition, they participate in shared biological pathways and molecular processes. The network topology indicates that proteins encoded by disease-associated genes are connected through common signalling hubs and pathways. These network-level associations suggest potential convergence of biological processes across conditions, but do not by themselves establish causal regulatory relationships. This convergence is particularly evident in regions of high node density, where interactions between genes from different conditions are concentrated.

To quantify network interconnectivity, degree centrality metrics were applied to the merged network. Figure 3C displays the top 50 most interconnected nodes, with colour intensity representing the degree of connectivity (red indicating the highest connectivity). Notably, while the majority of these top 50 hub genes—such

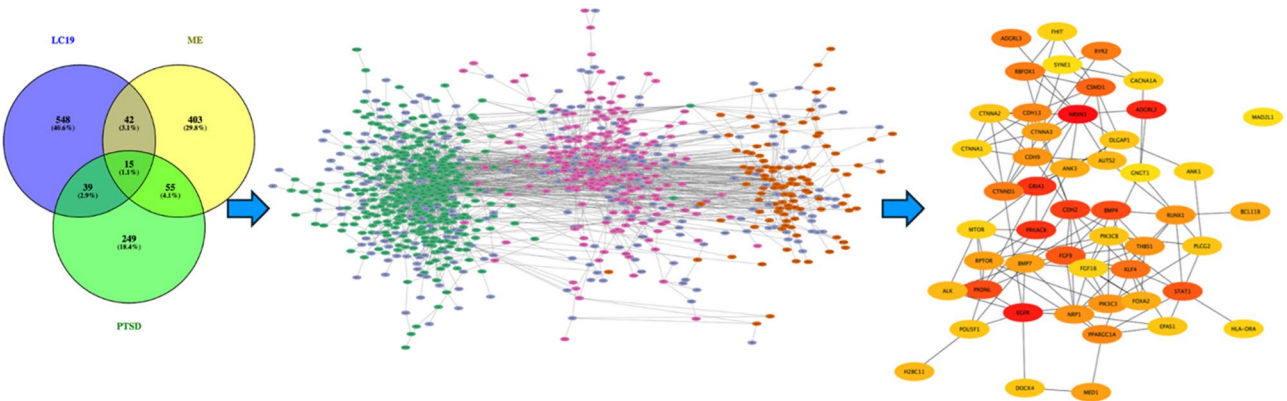


Fig. 3 Limited Gene-Level Overlap Across ME/CFS, Long COVID, and PTSD. **(A)** VENN Diagram: Shows the comparison of the EpiSwitch GWAS anchors for Long COVID, ME/CFS, and PTSD (Table 1) after they have been mapped to the closest coding region. It illustrates that the majority of genes identified in each condition are unique, with low gene-level overlap and only a small subset shared across two or more diseases. Note: Venn diagram overlaps **(A)** are calculated at the mapped gene level rather than raw genomic anchor coordinates to provide a unified basis for comparing disease-associated loci. **(B)** Merged STRING Network: Represents the merged STRING protein–protein interaction networks for Long COVID, ME/CFS, and PTSD gene sets. The nodes are spatially separated by gene set and by colour. Unlike the VENN diagram, the genes/proteins in each disease set exhibit high interconnectivity and a dense structure. **(C)** Top Connected Nodes: Highlights the top 50 nodes based on the degree of connection within the merged network. Nodes are coloured by their degree of connection, with red indicating the highest connectivity. Representative network visualisations; complete gene sets are provided in Supplementary Table 4

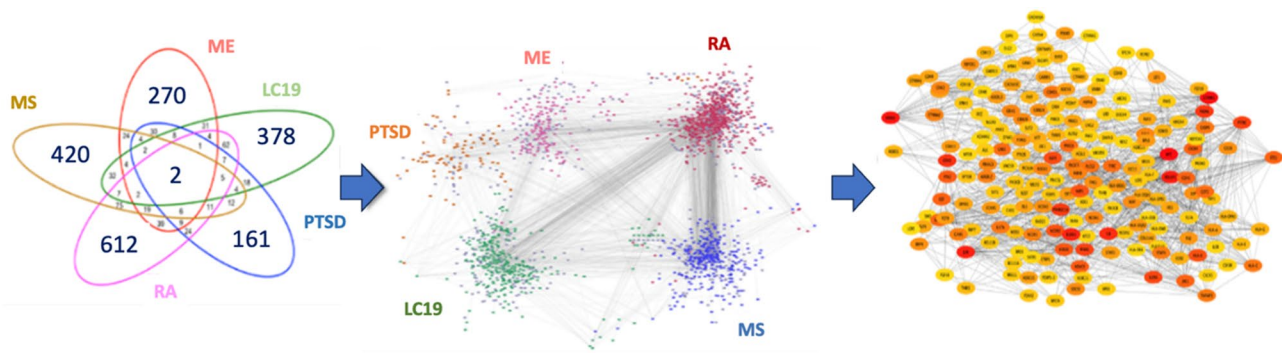


Fig. 4 Expansion of multi-disease network analysis to include RA and MS. **(A)** The Venn diagram summarises the gene overlap among Orion GWAS datasets after anchor-to-gene mapping. It displays the overlap of anchors across five Orion GWAS datasets, where the anchors were defined by comparing each disease-specific anchor set to the reference genome. Note: Cross-condition overlaps (A) represent shared mapped target genes across disease sets, reflecting shared functional pathways rather than identical base-pair anchor coordinates. **(B)** 5 EpiSwitch GWAS Anchors STRING networks were created for Long Covid (LC), PTSD, ME/CFS, Rheumatoid Arthritis (RA), & Multiple Sclerosis (MS) gene sets (RA: 965 anchors, 885 genes; MS: 730 anchors, 629 genes; LC: 611 anchors, 567 genes; ME/CFS: 552 anchors, 567 genes; PTSD 362 anchors, 324 genes). The networks and their overlaps are visualized in Cytoscape, with the nodes spatially separated by gene set and by colour. The resulting STRING networks reveal extensive interconnections among the conditions, forming a continuous multi-disease interaction landscape. **(C)** The top 200 interconnected nodes are extracted and coloured by degree of connectivity, with red indicating the highest connectivity. Representative network visualisations; complete gene sets are provided in Supplementary Table 5

as STAT1, PIK3CB, and EGFR—originate from a single condition dataset (predominantly Long COVID), they serve as vital bridges linking genes across multiple disease datasets within the broader network. Furthermore, a distinct subset of these highly connected central nodes, including ADGRL2, AUTS2, RPTOR, DOCK4, ALK, and ADGRL3, directly overlaps between at least two conditions. The spatial distribution of these high-degree nodes reveals a core regulatory backbone shared across ME/CFS, Long COVID, and PTSD. These highly connected nodes may represent candidate regulatory hubs within disease-relevant biological networks. Their high network connectivity suggests potential involvement in shared biological processes, but does not establish a causal role in disease pathogenesis. Across all analyses, a consistent pattern emerged: while gene-level overlap between conditions was limited, higher-order network organisation revealed extensive interconnectivity.

Expansion of multi-disease network analysis using the orion platform

Following the initial EpiSwitch® Orion and STRING analyses across ME/CFS, Long COVID, and PTSD, we expanded the same systems biology to include two additional immune-mediated conditions: multiple sclerosis (MS) and rheumatoid arthritis (RA).

Figure 4 illustrates the progressive expansion of this multi-disease model. Figure 4A presents a five-way Venn diagram comparing genes mapped from disease-associated anchors across ME/CFS, Long COVID (LC19), PTSD, MS, and RA. Each condition retained a substantial number of unique genes, reflecting the heterogeneous and polygenic nature of these disorders. RA and MS showed particularly large disease-specific gene sets,

while ME/CFS, Long COVID, and PTSD also demonstrated predominantly distinct genomic signatures. Only a small number of genes were shared across all conditions, confirming that direct gene overlap is limited even among clinically or biologically related disorders.

Using EpiSwitch® Orion, disease-associated loci were transformed into 3D genomic anchors, enabling non-coding variants to be linked to their likely functional gene targets rather than nearest linear genes alone. This substantially increased interpretive resolution and allowed incorporation of population-scale genetic evidence into the previously established EpiSwitch® biomarker framework. While the original analyses demonstrated limited direct gene overlap between ME/CFS, Long COVID, and PTSD (Fig. 4A), Fig. 4B visually demonstrates this convergence at the network level, with genes from each disease, shown in different colours, displaying extensive interconnectivity and forming a continuous multi-disease interaction landscape. The accompanying pathway analyses further indicate convergence across shared biological processes and signalling pathways. EpiSwitch® Orion was used to extend and refine this model across a broader spectrum of chronic inflammatory disease by integrating GWAS-derived signals, curated genomic resources, and three-dimensional chromatin interaction mapping. When the disease-specific gene sets were integrated into a common interaction network, a markedly different pattern emerged. Genes initially clustered according to disease source (ME/CFS (pink), Long COVID (green), PTSD (orange), MS (blue), and RA (red)) are connected by extensive cross-linking edges forming a continuous multi-disease interaction landscape. RA and MS formed particularly dense subnetworks consistent with their recognised autoimmune biology, while ME/CFS and Long

COVID connected strongly into these inflammatory hubs.

To identify shared drivers of disease biology, highly connected nodes were extracted from the combined network. Figure 4C displays the top hub genes ranked by degree centrality, with red nodes representing the most interconnected elements. Based on network topology, the most important identified hub genes driving these shared disease pathways included RUNX1, CDH2, PPARGC1A, NRP1, PLCG2, PIK3C3, and CSMD1. These hubs formed a dense shared core enriched for immune regulation, cytokine signalling, antigen presentation, chemokine activity, stress-response pathways, and mitochondrial/metabolic regulation. Prominent HLA genes (such as the highly centralised HLA-DQA2), STAT family members (such as STAT1), inflammatory mediators, and signalling adaptors (such as PLCG2) suggest that adaptive immunity and inflammatory regulation are prominent features of the integrated network. Furthermore, the high connectivity of metabolic and transcriptional regulators such as PPARGC1A and RUNX1 highlights candidate regulatory nodes associated with cellular energy management and developmental pathways within this multi-disease network. These network-level associations should be considered predictive and hypothesis-generating rather than evidence of causal involvement.

Importantly, many of the highest-ranked hubs were not the strongest direct GWAS hits, but genes occupying strategic regulatory positions within the network. This emphasises the added value of EpiSwitch[®] Orion compared with conventional linear association studies, as biologically important genes may emerge through connectivity rather than locus significance alone.

Overall, inclusion of MS and RA substantially strengthened the original EpiSwitch[®] findings by demonstrating that ME/CFS, Long COVID, PTSD, and established autoimmune diseases converge on overlapping regulatory systems despite retaining distinct genetic signatures. These data support the concept that chronic complex illnesses may be best understood not as isolated disease entities, but as related perturbations of shared three-dimensional genomic and immunoregulatory networks.

Integration Across Diseases: Convergence Without Overlap

Taken together, the results demonstrate a consistent pattern across all analyses:

1. **Low gene-level overlap** between ME/CFS, Long COVID, and PTSD.
2. **High network-level interconnectivity**, indicating shared functional pathways.
3. **Identification of central hub genes** that bridge disease-specific networks.

4. **Convergence on key biological processes**, including immune, metabolic, and neurological pathways.

This combination of findings supports a model in which complex chronic diseases are driven by perturbations in shared regulatory networks rather than identical gene sets. The use of 3D genomic data is critical for revealing these relationships, as it captures spatial and regulatory interactions that are invisible to linear genomic analyses.

Overall, the integration of EpiSwitch[®] platforms provides a comprehensive view of disease biology, bridging the gap between genetic variation and functional outcome.

Discussion

The present study demonstrates that ME/CFS, Long COVID, PTSD, multiple sclerosis (MS), and rheumatoid arthritis (RA) exhibit limited overlap at the level of individual genes but converge strongly at the level of three-dimensional (3D) genomic regulatory networks. This apparent paradox—minimal gene overlap across these five diverse inflammatory and immune-mediated conditions despite substantial functional convergence—aligns with and extends current understanding of these disorders as complex, multisystem conditions driven by dysregulated biological networks rather than single-gene effects. By integrating EpiSwitch[®] chromosome conformation signatures and EpiSwitch[®] Orion disease-specific anchor landscapes through network biology, our findings provide a mechanistic framework that bridges prior clinical and molecular observations across this broader spectrum of chronic disease.

Reconciling low genetic overlap with shared disease biology

A key observation from this study is the discordance between gene-level overlap and network-level convergence. While EpiSwitch[®] Orion GWAS-derived gene sets for ME/CFS, Long COVID, and PTSD showed minimal direct overlap (Fig. 3A), network integration revealed extensive interconnectivity (Fig. 3B). This finding is highly consistent with emerging literature indicating that complex chronic diseases are polygenic and heterogeneous, with risk distributed across many loci of small effect rather than shared core genes [19, 20, 28, 29, 37, 38].

Large-scale genetic studies, including the DecodeME initiative, have identified only a small number of reproducible loci associated with ME/CFS, and notably, these loci do not substantially overlap with those identified in Long COVID [65]. Similarly, clinical and genetic studies have emphasised that ME/CFS is a highly heterogeneous disorder with multiple contributing pathways rather than

a single unifying genetic cause [1]. The absence of strong gene overlap between ME/CFS and Long COVID, despite striking clinical similarities, has been interpreted as evidence of distinct etiological triggers acting on shared biological systems.

Our findings provide a resolution to this apparent contradiction. By incorporating 3D genomic architecture, we demonstrate that distinct gene sets can converge onto shared regulatory networks. Specifically, the emergence of high-degree hub genes such as *RUNX1*, *CDH2*, and *NRP1* across our five-disease expansion identifies a candidate shared regulatory network associated with systemic inflammatory processes. These highly connected nodes may contribute to the observed network-level convergence, although their functional and causal roles remain to be established experimentally. This suggests that disease similarity arises not from identical genes, but from the organisation of those genes within common interaction frameworks. In other words, different genetic inputs may perturb the same regulatory circuits, producing similar phenotypic outcomes.

This network-centric view is increasingly supported in systems biology, where disease phenotypes are understood as emergent properties of dysregulated networks rather than isolated molecular defects. The EpiSwitch® Orion platform, by integrating 3D genomic landscapes with protein–protein interaction networks and pathway databases, provides a powerful tool to uncover these hidden relationships.

Integration with clinical overlap between ME/CFS and long COVID

The convergence observed in our network analyses mirrors extensive clinical overlap between ME/CFS and Long COVID. Systematic reviews have shown that the majority of ME/CFS symptoms—up to 25 out of 29—are also reported in Long COVID cohorts [66]. These include fatigue, cognitive dysfunction (“brain fog”), sleep disturbances, and autonomic dysfunction, all of which are characteristic features of both conditions. Moreover, it has been estimated that a substantial proportion of Long COVID patients meet diagnostic criteria for ME/CFS, particularly in post-infectious cohorts. This reinforces the concept that Long COVID may, in part, represent a post-viral trigger for ME/CFS-like pathology. However, important differences remain, including organ-specific sequelae in Long COVID related to acute viral injury.

Our findings suggest that these similarities and differences can be explained through shared network-level biology combined with disease-specific perturbations. The low gene overlap observed in Fig. 3A is consistent with the lack of shared GWAS loci between ME/CFS and Long COVID. However, the strong network interconnectivity observed in Fig. 4 indicates that both conditions

converge on similar biological pathways, including PDGFR-beta signalling, CXCR4-mediated signalling, and the glucocorticoid receptor network. This supports a model in which different initiating events lead to dysregulation of a common set of interconnected biological systems. The 3D genome provides a mechanistic substrate for this convergence, as spatial chromatin interactions enable coordinated regulation of genes across the genome.

Immune dysregulation and T-cell exhaustion as central mechanisms

One of the most prominent themes emerging from both our results and the literature is the role of immune dysregulation. Pathway enrichment analysis (Fig. 2B) highlighted cytokine signalling, interferon pathways, and immune activation as central features of ME/CFS networks. These findings are consistent with multiple studies demonstrating altered immune function in both ME/CFS and Long COVID.

Recent transcriptomic analyses have identified signatures of immune exhaustion in both conditions, including downregulation of interferon signalling and impaired macrophage activation [67]. These findings suggest a state of chronic immune activation followed by functional exhaustion, which may contribute to persistent symptoms. Our network analysis identified *STAT1* and *HLA-DQA2* as central hubs, reinforcing the importance of the adaptive immune response and antigen presentation in maintaining this chronic inflammatory state.

LAG3 emerged as a notable node within the ME/CFS network (Fig. 2D), showing functional connectivity with established ME/CFS-associated genes and immune regulatory pathways. LAG3 (lymphocyte activation gene 3) is an immune checkpoint molecule associated with T-cell exhaustion and impaired T-cell function during chronic antigenic stimulation [68]. Its positioning within the ME/CFS network therefore provides a potential molecular link between chronic immune activation and altered T-cell regulation. The identification of LAG3 alongside other immune signalling components, including *STAT1*, further supports the hypothesis that persistent immune activation and dysregulated immune regulation may contribute to the biological architecture of ME/CFS.

This interpretation is consistent with independent evidence linking persistent viral infection with immune dysregulation in ME/CFS. Rasa-Dzelzkaleja et al. [17] reported persistent HHV-6 A/B, HHV-7, and parvovirus B19 infection/co-infection in ME/CFS, with active infection associated with increased levels of pro-inflammatory cytokines, including IL-6, TNF- α , and IL-12. These findings provide biological context for a model in which persistent antigenic or inflammatory stimulation may contribute to sustained immune activation and T-cell

dysfunction. Our network analysis does not establish that LAG3-mediated T-cell exhaustion is a causal mechanism in ME/CFS; rather, LAG3 represents a candidate immune-regulatory node whose network position is consistent with this broader hypothesis and warrants independent functional investigation.

Mitochondrial dysfunction and metabolic reprogramming

Another key area of convergence identified in this study is mitochondrial function and metabolic regulation. Pathway analysis revealed enrichment of genes involved in mitochondrial biogenesis, glycolysis, and fatty acid metabolism within ME/CFS networks (Fig. 2B). The identification of *PPARGC1A* (PGC-1 α) as a top-tier hub gene provides a significant mechanistic link. *PPARGC1A* is the master regulator of mitochondrial biogenesis and its central role in our network suggests that metabolic “exhaustion” is genetically encoded through 3D regulatory anchors in these conditions. Emerging multi-omics studies have further demonstrated disruptions in metabolic pathways, including altered mitochondrial function and shifts in energy utilization. For example, integrative analyses have shown coordinated changes across these systems in ME/CFS, highlighting the interconnected nature of metabolic and immune processes [21]. Long COVID has similarly been associated with metabolic dysfunction, including impaired mitochondrial activity and altered energy homeostasis. These shared features support the concept of a “metabolic trap” or persistent energy deficit underlying fatigue and post-exertional malaise in both conditions. The high connectivity of *PIK3C3*, involved in autophagy and nutrient signalling, further supports the conclusion that cellular “recycling” and energy sensing are fundamentally perturbed.

Neuroendocrine and neuroimmune integration

The inclusion of PTSD in this analysis provides an important perspective on the neuroendocrine and neuroimmune dimensions of these conditions. PTSD has traditionally been conceptualised as a psychiatric disorder, but increasing evidence points to systemic biological changes, including alterations in immune function, stress hormone regulation, and neural connectivity.

Our network analysis demonstrates that genes associated with PTSD are embedded within the same interconnected networks as those associated with ME/CFS and Long COVID. Notably, the presence of *CDH2* (Cadherin 2) as a primary hub points to a shared role for synaptic adhesion and neural plasticity across these syndromes. This supports a unified model in which stress-related and immune-related pathways interact within a shared regulatory framework. The recurrence of the Glucocorticoid Receptor signalling and ATF-2 transcription networks across our analyses suggests a common mechanism for

how chronic physiological or psychological stress is translated into systemic immune dysregulation.

Diagnostic performance and implications for precision medicine

The previously reported 96% diagnostic accuracy of the EpiSwitch® ME/CFS classifier, established using an independent validation cohort [58], provides evidence supporting the diagnostic potential of chromosome conformation signatures. However, this diagnostic validation should be distinguished from the network analyses performed in the present study, which identify candidate hub genes and shared regulatory pathways but do not independently validate their mechanistic significance. By focusing on the “regulatory control room” (the 3D genome) rather than just the “output” (mRNA or proteins), we can identify stable markers that reflect the underlying pathology of complex chronic diseases.

This is particularly important in the context of ME/CFS and Long COVID, where diagnosis is currently based on clinical criteria and exclusion. The involvement of hub genes like *RUNX1* (involved in haematopoiesis and immune regulation) and *PLCG2* (signalling adaptor) suggests that these biomarkers are tracking the most vital regulatory points in the disease network, rather than peripheral noise.

Distinction between diagnostic validation and Network-based gene prioritisation

The diagnostic performance reported in this study is based on the previously published EpiSwitch® ME/CFS biomarker study [58], rather than on the present multi-disease network analysis. In that study, a 200-marker chromosome conformation classifier was developed in a discovery cohort and subsequently evaluated in an independent validation cohort comprising 24 ME/CFS cases and 45 controls, achieving 92% sensitivity, 98% specificity and 96% overall accuracy [58]. These validation samples were not used during classifier training.

The network analysis presented here addresses a different question. Rather than independently testing diagnostic performance, it integrates the previously characterised ME/CFS biomarker landscape with EpiSwitch® Orion-derived GWAS anchor datasets from ME/CFS, Long COVID, PTSD, MS and RA. Genes mapped from these datasets were analysed using protein–protein interaction networks, and highly connected nodes were prioritised according to network topology. Accordingly, genes such as *RUNX1*, *PPARGC1A*, *CDH2*, *NRP1* and *PLCG2* represent candidate hub genes emerging from systems-level network analysis. Their identification is hypothesis-generating and does not constitute independent experimental validation of their mechanistic role in ME/CFS or the other conditions studied.

Thus, the previously demonstrated diagnostic validation of the EpiSwitch® ME/CFS classifier and the network-based identification of candidate regulatory hubs should be considered complementary but distinct findings. Independent validation of the predicted hub genes in appropriately characterised clinical cohorts, together with functional experimental studies, will be required to establish their mechanistic significance.

A systems biology model of chronic disease

Our findings support a systems biology model in which ME/CFS, Long COVID, PTSD, MS, and RA represent different manifestations of dysregulated biological networks. In this model, while the initial “insult” varies, the downstream network perturbations converge on a shared core of immune and metabolic regulators—most notably the *RUNX1-PPARGC1A-STAT1* axis.

This framework aligns with recent advances in multi-omics research, which emphasise the importance of integrating data across different biological layers to understand complex diseases. Variability in genetic background and environmental exposures lead to different disease trajectories, but commonality of these 3D genomic hubs provides a rationale for investigating whether modulation of shared regulatory pathways could have therapeutic relevance across multiple chronic inflammatory conditions. However, whether targeting these candidate nodes can produce therapeutic benefit, and whether such effects would be shared across diseases, remains to be established through functional experiments, preclinical studies, and appropriately designed clinical trials.

Study limitations

This study has several limitations. **First**, the analyses rely on the integration of heterogeneous GWAS datasets that differ in cohort composition, sample size, ancestral breakdown, and phenotypic definitions (Table 1). Because this framework utilizes publicly available GWAS summary statistics rather than raw individual-level genotype data, control for primary population stratification, ancestry, and standard covariates is inherited directly from each original GWAS association-testing framework (e.g., principal-component-adjusted regression and genomic control). While standard for downstream summary-statistic analyses (e.g., Mendelian randomization, LDSC, and polygenic risk scoring), this approach cannot account for unmeasured environmental exposures or variations in covariate adjustment schemes across source studies. Furthermore, cross-study heterogeneity in sample sizes and statistical power directly influences 3D anchor discovery density, which accounts for variations in the total number of genomic anchors and mapped target genes identified across conditions.

Second, a formal permutation-based statistical validation of network convergence against a random null model was not performed. Protein–protein interaction networks were generated in STRING under default settings, which prioritize high-confidence, experimentally validated interaction evidence rather than lower-confidence channels (e.g., text-mining or co-expression alone). Cytoscape was used solely to merge and visualize these pre-validated networks without introducing additional edges or altering the underlying evidence base. This framework differs from network-propagation methods (such as random walk with restart), where computational connectivity propagation introduces susceptibility to hub-gene topology bias and necessitates permutation testing. Because STRING edges under default settings reflect independently validated biological interactions rather than topological artifacts, the observed cross-condition convergence likely reflects shared biology; nonetheless, degree-matched permutation testing remains a valuable direction for future validation.

Third, the network analyses identify candidate regulatory convergence but do not establish causal biological relationships or provide direct functional experimental validation of predicted shared hubs across independent disease cohorts. Notably, while the independent ME/CFS validation cohort described in Hunter et al. (2025) [58] validates the diagnostic performance of the previously developed 200-marker EpiSwitch® classifier, it does not independently validate the hub genes identified through the present multi-disease network analysis. These candidate hubs—including *RUNX1*, *PPARGC1A*, *CDH2*, *NRP1*, and *PLCG2*—were prioritized based on network topology following the integration of EpiSwitch® Orion-derived GWAS anchors and interaction networks. Consequently, these findings should be interpreted as hypothesis-generating, requiring independent replication in external cohorts and functional assays to confirm their mechanistic driver roles.

Fourth, while the EpiSwitch platform captures dynamic chromatin architecture, longitudinal studies are required to determine the temporal stability of these 3D genomic signatures over disease course and treatment.

Fifth, the present analysis is disease-centric rather than directly phenotype-centric. The input GWAS and EpiSwitch® datasets are structured around diagnostic categories, lacking a harmonized multi-disease cohort stratified by fatigue presence, severity, or duration. Consequently, cross-disease network convergence cannot be definitively attributed specifically to fatigue itself, as shared biology may reflect broader overlapping systemic mechanisms, such as immune activation, systemic inflammation, or metabolic dysregulation. A definitive phenotype-centric analysis will require multi-diagnostic cohorts with harmonized clinical fatigue instruments

directly incorporated into genomic workflows, for which the present findings provide a hypothesis-generating foundation.

Conclusion

Overall, this study identifies candidate areas of regulatory convergence across ME/CFS, Long COVID, PTSD, multiple sclerosis and rheumatoid arthritis using integrated 3D genomic and network-based analyses, despite minimal overlap in gene identity. The identification of key hub genes such as RUNX1, PPARGC1A, and CDH2 provides a new set of targets for understanding how different triggers converge on the same clinical exhaustion. By integrating 3D genomic anchors with network biology, we provide a unifying framework that explains both the similarities and differences between these conditions. The identification of highly connected genes and shared pathways provides hypotheses for further investigation of disease-associated regulatory architecture and potential candidate biomarkers and therapeutic targets. Further investigations and clinical trials are required to establish clinical diagnostic utility, causal biological mechanisms, or therapeutic efficacy.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12967-026-08874-9>.

Supplementary Material 1: Supplementary Table 1. STRING Network for top 200 ME/CFS Biomarker Genes (Fig. 2A).

Supplementary Material 2: Supplementary Table 2. Pathway datasets for Reactome, NCI and Single Cell Marker pathways of ME/CFS Biomarker Genes (Fig. 2B, C).

Supplementary Material 3: Supplementary Table 3. Top predicted nodes for pathways of ME/CFS Biomarker Genes (Fig. 2D).

Supplementary Material 4: Supplementary Table 4. Pathway and nodes datasets for STRING Network analysis for Overlap Across ME/CFS, Long COVID, and PTSD (Fig. 3B, C).

Supplementary Material 5: Supplementary Table 5. Gene and pathway datasets for STRING Network analysis for Multi-Disease Network Analysis (Fig. 4B, C).

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Author contributions

EH, AA and DP conceived the study. KC supervised clinical sample collection, provided clinical insight into the cohort stratification and reviewed the manuscript. DV, SB, AG, AV, JC, AD, RP, MS, JG, conducted clinical sample preparation and analysis. EH, HA, AA and DP analysed data, wrote and reviewed the manuscript.

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Data availability

Knowledge graphs were generated using the EpiSwitch 3D-Genomics large language model (LLM), which employs a unique embedding approach. This platform integrates advanced semantic parsing with over 1.5 billion experimentally-verified 3D genome interactions. Leveraging proprietary AI models trained on chromatin architecture, it enables mapping of regulatory circuitry, identification of causal mechanisms, and prediction of the effects of non-coding and structural variants within their native 3D genomic context. These tools are deployed on Google Cloud in collaboration with Google. Data is available on request <https://www.oxfordbiodynamics.com/contact-us#contact-us-form-wrapper>.

Declarations

Consent for publication

Written informed consent for publication was obtained from all authors.

Competing interest

EH, DV, SB, AG, AV, JC, AD, RP, MS, JG, and AA are full-time employees at Oxford BioDynamics plc and have no other competing financial or other interests. None of the remaining authors has competing interests.

Ethical consent and guidelines

All ethical guidelines were followed.

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